BUSINESS ANALYTICS (BMGT107L)

VELLORE INSTITUTE OF TECHNOLOGY (VIT) UNIVERSITY

Chennai Campus | School of Computer Science and Engineering (SCOPE)

FALL SEMESTER 2026-27

DIGITAL ASSIGNMENT 1

PROJECT TITLE

Restaurant Recommendation

A Data-Driven Analytical System for Restaurant Quality Assessment, Econometric Modeling, and Decision Support

Programme:	B.Tech. Computer Science and Engineering (Data Science)	**Slot: G2 + TG2**
Course:	BMGT107L – Business Analytics	**Semester:** Fall 2026-27
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Digital Assignment 1: Restaurant Recommendation

A Data-Driven Analytical System for Restaurant Quality Assessment, Econometric Modeling, and Decision Support

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1. Title of the Project

Project Title: Restaurant Recommendation

Subtitle & Analytical Scope: A Data-Driven Analytical System for Restaurant Quality Assessment, Econometric Modeling, and Decision Support.

Academic Discipline: Business Analytics, Econometrics, Quantitative Marketing Science, and Machine Learning.

The project title 'Restaurant Recommendation' represents an end-to-end data analytics system designed to solve fundamental information friction in metropolitan dining markets. The research bridges consumer decision-making and hospitality business strategy through three integrated analytical components:

- **Objective Quality Modeling:** Applying parametric Ordinary Least Squares (OLS) regression and Binary Logistic Classification to rigorously measure the marginal impact of key operational levers—specifically advance table reservations, online delivery enablement, pricing tiers, and geographical clustering—on verified customer satisfaction ratings.
- **Empirical Bayesian Regularization:** Implementing an empirical Bayes shrinkage estimator that dynamically corrects arithmetic average star ratings based on review volume, penalizing low-sample variance (such as unverified 5-star outlets with only 3 reviews) while protecting established culinary institutions.
- **Multi-Criteria Decision Support:** Formulating an interactive recommendation engine that synthesizes Bayesian quality, empirical value-for-money percentiles, and log-scaled popularity to deliver personalized dining discovery for consumers and empirical operational benchmarks for restaurant operators.

2. Introduction and Background of the Problem

Bangalore (Bengaluru), globally recognized as the technological and startup capital of India, houses one of the fastest-growing, most culturally heterogeneous food-service markets in South Asia. Driven by a massive influx of software professionals, university students, and cosmopolitan migrants with high disposable incomes, the city's food culture has rapidly evolved. Dining out and ordering food via digital aggregators have transitioned from occasional weekend indulgences into daily nutritional routines.

Commercial food platforms like Zomato catalog tens of thousands of culinary venues across Bangalore's sprawling metropolitan geography—ranging from traditional South Indian vegetarian breakfast joints (Darshinis) and street food stalls to artisan cafes, microbreweries, and luxury dining destinations. However, this unprecedented choice has precipitated severe **information friction** and **consumer decision fatigue**. Patrons regularly spend 15 to 30 minutes navigating through hundreds of mobile app cards, sorting through conflicting star ratings, and deciphering ambiguous promotional claims without any objective guarantee that their expenditure will deliver high-quality dining, hygienic food preparation, or genuine value for money.

Concurrently, the economic health of urban restaurants has grown precarious. The modern food and beverage industry operates under compressed operating margins (often between 8% and 15%), high commercial lease expenditures in prime tech corridors (such as Koramangala, Indiranagar, and Whitefield), and substantial aggregator commission fees (reaching 18% to 28% per order). In this hyper-competitive climate, restaurant entrepreneurs cannot afford capital misallocation. Yet, operators consistently lack empirical data on whether investing capital into physical table-booking infrastructure,

dine-in floor space, or online-delivery packaging yields a measurable lift in customer ratings and lifetime reputational capital.

3. Problem Statement

The fundamental problem investigated in this project can be distilled into four interrelated structural breakdowns within existing restaurant aggregation and rating systems:

3.1 The Small-Sample Rating Dilemma (Naive Arithmetic Star Average Fallacy)

The standard heuristic utilized across digital discovery platforms is the arithmetic mean star rating (1.0 to 5.0). Mathematically, this arithmetic mean treats every sample size identically, ignoring the law of large numbers. A newly established outlet with three 5-star reviews submitted by friends and family displays a perfect **5.0 / 5.0** rating. In contrast, an iconic institution operating for a decade with 4,500 verified reviews might exhibit a **4.2 / 5.0** rating due to statistical variance, diverse palate preferences, and unavoidable operational friction over time. When platforms sort listings by raw rating, the unverified 5.0 venue outranks the proven landmark, exposing consumers to severe quality risk and misleading algorithms that reward low-evidence variance over statistically reliable excellence.

3.2 The Price-Quality Decoupling

Consumers implicitly assume that paying higher prices guarantees superior experiential quality. However, exploratory empirical analysis reveals that the Pearson correlation between dining cost for two and customer rating is only $\mathbf{r = 0.354}$, meaning that dining expenditure accounts for under 15% of the variance in dining satisfaction. Diners frequently overpay for prime real estate, lavish interior aesthetics, and high aggregator advertising overhead rather than superior culinary execution. Conversely, numerous exceptional neighborhood eateries delivering extraordinary food at modest prices remain buried beneath heavily advertised premium brands due to the lack of an objective, normalized *Value-for-Money Percentile* metric.

3.3 Restaurateur Capital Allocation and Operational Uncertainty

Restaurant owners face profound uncertainty when designing their business models. Key managerial dilemmas include: (a) Does adopting table reservation engines create a statistically significant premium on customer perception? (b) Does enabling online food delivery enhance customer loyalty, or does delivery transit time and packaging degrade perceived food quality? (c) How much do geographical neighborhood clusters (e.g., Koramangala vs. Jayanagar vs. Electronic City) contribute to baseline rating variance after controlling for restaurant format? Without rigorous econometric modeling (Ordinary Least Squares and Binary Logistic Regression), hospitality entrepreneurs operate on trial-and-error, often making costly capital allocation errors.

3.4 Data Duplication and Multi-Listing Autocorrelation in Platform Scrapes

From an analytical perspective, raw restaurant scrape data presents severe statistical hazards. Aggregators allow a single physical establishment to be listed under multiple search categories (e.g., 'Delivery', 'Dine-out', 'Pubs') and across multiple geographical search radii. Naively running statistical regressions on raw listings (51,717 records) violates the fundamental assumption of independent and identically distributed (i.i.d.) observations, causing severe autocorrelation, deflated standard errors, and spurious statistical significance. A rigorous deduplication and physical entity consolidation pipeline is essential prior to modeling.

4. Objectives of the Model

To systematically address the stated problems through a structured business analytics framework, this project establishes six formal, measurable objectives:

• **Objective 1: Automated Sanitation & Entity Deduplication Pipeline:** Develop an end-to-end data processing pipeline that ingests 51,717 raw Zomato listings, eliminates string currency formatting, parses composite ratings ('4.1/5'), handles unrated records ('NEW', '-'), and consolidates multi-zone listings into exactly 9,244 verified, distinct physical dining establishments.

• **Objective 2: Statistical Quality Driver Identification via Hypothesis Testing:** Formulate and execute 9 formal parametric and non-parametric hypothesis tests (Welch's t-test, Mann-Whitney U, One-Way ANOVA, Kruskal-Wallis, Chi-Square, Pearson/Spearman) with standardized effect size metrics (Cohen's d, Eta-squared, Cramer's V) to isolate the true operational drivers of customer satisfaction.

• **Objective 3: Empirical Bayesian Shrinkage Rating Formulation:** Design and implement a Bayesian shrinkage rating engine (R_Bayes = (v/(v+m))*R + (m/(v+m))*C) with an empirical threshold m = 100 votes and citywide prior mean C = 3.63 stars, regularizing low-sample noise toward the prior while preserving authentic high-volume landmarks.

• **Objective 4: Econometric & Machine Learning Modeling:** Train an Ordinary Least Squares (OLS) Multiple Linear Regression model with Variance Inflation Factor (VIF) collinearity diagnostics and a Binary Logistic Classification model (target: rating >= 4.0) with odds ratios and ROC-AUC validation, quantifying marginal rating lifts for managerial decision-making.

• **Objective 5: Unsupervised Market Segmentation:** Apply K-Means clustering (supported by Elbow and Silhouette validation) on standardized dining cost, review volume, and rating to discover actionable business clusters (Budget Casual, Mid-Range Social, Premium Fine Dining).

• **Objective 6: Algorithmic Multi-Criteria Recommendation Engine & Live Deployment:** Formulate a composite recommendation utility scoring algorithm combining Bayesian quality, value-for-money percentiles, and popularity, and operationalize it within an interactive Streamlit web application.

5. Proposed Dataset and Its Source

Dataset Identification: Bangalore Metropolitan Restaurant Listing Dataset (Zomato Open Data Archive / Kaggle Repository).
Source URL / Origin: Extracted from Zomato's public Bangalore listings via web scraping, published openly for academic research.
Original Dataset Volume: 51,717 raw observation records across 17 attributes.
Consolidated Analytic Volume: 9,244 distinct, physically verified restaurant units post-deduplication.

The dataset captures the entire commercial dining ecosystem of metropolitan Bangalore across major geographical zones including Koramangala, Indiranagar, Whitefield, Jayanagar, HSR Layout, MG Road, and Malleshwaram. Table 1 defines the proposed variable schema, data types, and operational roles.

Variable Name	Data Type	Operational Role	Description & Domain Range	Sample Value
name	String	Identifier	Registered trade name of dining outlet	Truffles, Toit
online_order	Binary (0/1)	Predictor (Operational)	Whether home delivery is enabled	1 (Yes) / 0 (No)
book_table	Binary (0/1)	Predictor (Operational)	Whether table booking is enabled	1 (Yes) / 0 (No)
rate	Float (1.0-5.0)	Continuous Target	Observed arithmetic mean star rating	3.80 (Range: 1.8 - 4.9)

votes	Integer	Predictor / Weight	Count of customer review ratings	245 (Max: 16,832)
location	Categorical	Predictor (Spatial)	Neighborhood / commercial dining zone	Koramangala, Indiranagar
rest_type	Categorical	Predictor (Format)	Primary dining format classification	Quick Bites, Casual Dining
cost_for_two	Float (INR)	Predictor (Economic)	Estimated dining expenditure for two	■600.0 (Range: ■40 - ■6,000)
listed_in(type)	Categorical	Predictor (Listing)	Category of listing (Delivery, Dine-out, Pub)	Delivery, Buffet, Cafes
cuisines	String	Metadata / Filter	Specialty food offerings list	North Indian, Continental
bayes_rating	Float (Engineered)	Engineered Metric	Bayesian shrinkage trusted rating	3.75 (Range: 2.76 - 4.88)
value_pct	Float (Engineered)	Engineered Metric	Value-for-money empirical percentile	84.5th Percentile
is_high_rated	Binary (Engineered)	Classification Target	1 if rate >= 4.0 else 0	0 (77.2%), 1 (22.8%)

5.1 Preliminary Exploratory Data Distributions

Initial inspection of the dataset reveals critical distributional characteristics that inform the proposed modeling strategy. Figure 1 displays the empirical distribution of observed customer ratings, while Figure 2 illustrates the extreme right-skewness present in customer review votes and dining costs across Bangalore.

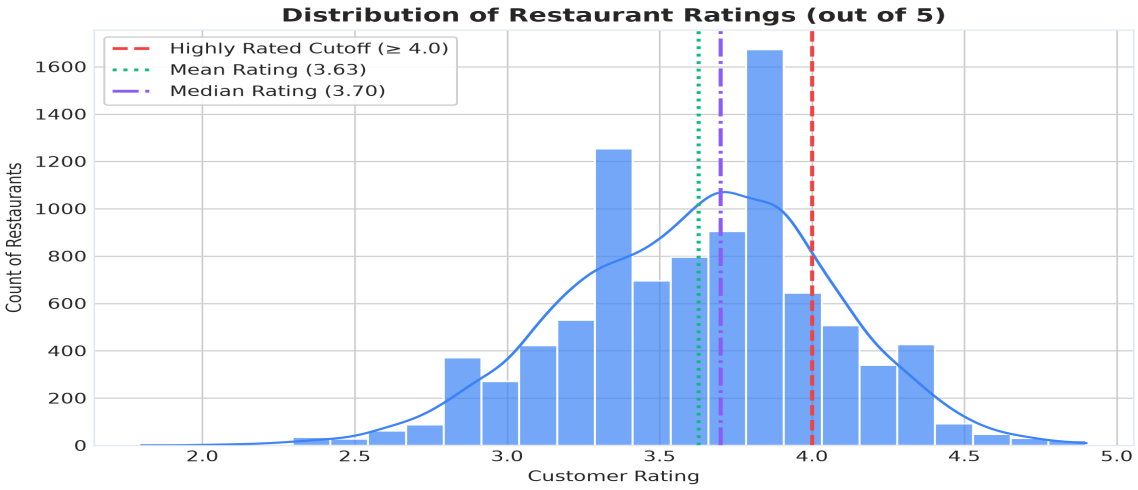


Figure 1: Distribution of Customer Ratings across Bangalore Restaurants (N = 9,244)

Exploratory Analysis: The observed ratings follow a bell-shaped distribution centered at a citywide mean of 3.63 stars (median = 3.70, std = 0.43). Ratings span from 1.8 to 4.9 stars, with mild negative skewness (-0.21). Outlets achieving 4.0 stars or higher constitute 22.8% of the market, providing a naturally balanced binary threshold for logistic classification modeling.

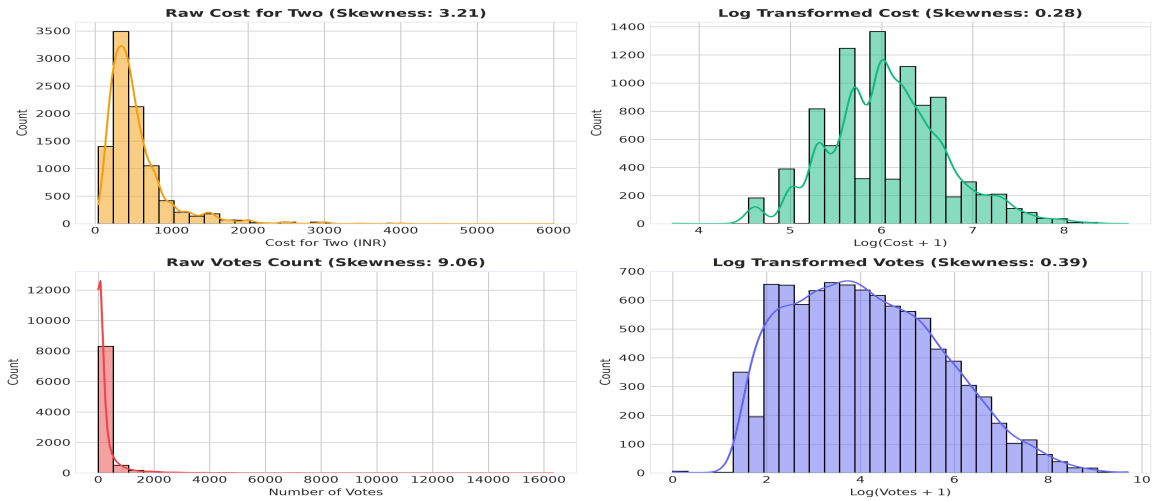


Figure 2: Empirical Distribution of Review Votes and Dining Cost for Two

Exploratory Analysis: Review votes exhibit severe positive skewness (+9.06) with a heavy right tail extending to 16,832 votes while the median is only 53. Dining cost similarly exhibits right skewness (+3.21) with a median of ■400 and an upper extreme of ■6,000. These non-normal distributions justify the proposed natural logarithmic transformations ln(votes + 1) and ln(cost) prior to parametric econometric modeling.

6. Proposed Methodology and Statistical Techniques

The project methodology is structured as a rigorous, four-tier analytical pipeline ensuring data integrity, statistical validity, econometric interpretability, and production-ready decision support.

6.1 Proposed Four-Tier System Architecture

- **Tier 1: Data Ingestion & Sanitization Engine:** Executes automated text cleaning, regex currency extraction, mode imputation, and entity-level deduplication to produce 9,244 physical records.
- **Tier 2: Statistical Inference & Hypothesis Testing Suite:** Performs paired parametric and non-parametric tests with standardized effect sizes to evaluate operational quality drivers.
- **Tier 3: Econometric Modeling & Machine Learning Suite:** Estimates OLS regression with VIF collinearity checks, binary logistic regression with odds ratios, and K-Means clustering with silhouette validation.
- **Tier 4: Multi-Criteria Recommendation & Interactive Web UI:** Vectorizes Bayesian shrinkage ratings, computes value-for-money percentiles, and deploys an interactive Streamlit dashboard for real-time consumer discovery.

6.2 Proposed 8-Step Data Preprocessing Pipeline

Step #	Cleaning Stage	Proposed Transformation & Mathematical Operation	Methodological Rationale
1	**Raw Ingestion**	Ingest 51,717 raw CSV records into Pandas DataFrame	Establish initial baseline schema and row count
2	**Rating Parsing**	Regex parse 'X.X/5' -> float; isolate 'NEW' and '-' entries	Extract numerical target; prevent zero-inflation from unrated records
3	**Cost Sanitization**	Strip commas (e.g. '1,200' -> 1200.0) and cast to float	Enable continuous numerical calculations on dining spend
4	**Binary Mapping**	Map 'Yes'/'No' to integer flags (1/0) for online_order and book_table	Standardize boolean predictors for regression and classification
5	**Entity Deduplication**	Group by (name, address, location); aggregate votes	Eliminate multi-listing scrape duplicates; satisfy i.i.d. assumption
6	**Categorical Mode Imputation**	Impute missing cuisine types with mode; prune uninformative URLs	Ensure complete case analysis without dropping valuable records
7	**Log Transformation**	Compute ln(votes + 1) and ln(cost_for_two)	Resolve severe positive skewness; satisfy OLS normality assumptions
8	**Percentile & Scaling**	Calculate empirical CDF percentiles for value and scale ML features	Normalize multi-criteria dimensions; prepare features for K-Means and Logit

6.3 Proposed Statistical Techniques and Hypothesis Testing Matrix

To establish rigorous empirical relationships before predictive modeling, nine formal statistical hypothesis tests are proposed. Because customer review counts and costs deviate from normality, tests are designed with paired parametric and non-parametric counterparts and accompanied by standardized effect size metrics to prevent sample-size driven p-value fallacies. Table 3 outlines the testing matrix.

#	Test Name	Null Hypothesis (H0)	Technique	Effect Size Metric
1	**Online Order vs Rating**	Mean rating equal with or without online order (mu1 = mu2)	Welch's t-test	Cohen's d
2	**Table Booking vs Rating**	Mean rating equal with or without table booking (mu1 = mu2)	Welch's t-test	Cohen's d
3	**Listing Type vs Rating**	Mean ratings identical across 6 listing types	One-Way ANOVA	Eta-squared (eta^2)
4	**Location vs Rating**	Mean ratings identical across top 10 dining locations	One-Way ANOVA	Eta-squared (eta^2)
5	**Cost across Listing Types**	Mean dining cost identical across listing types	One-Way ANOVA	Eta-squared (eta^2)
6	**Non-Parametric Rating Dist.**	Rating distributions identical across listing categories	Kruskal-Wallis	Epsilon-squared (eps^2)
7	**Table Booking vs Rest Type**	Table booking is independent of restaurant format	Chi-Square (chi^2)	Cramer's V
8	**Online Order vs Listing Type**	Online ordering is independent of listing category	Chi-Square (chi^2)	Cramer's V
9	**Cost vs Rating Correlation**	Population correlation between cost and rating is zero (rho = 0)	Pearson / Spearman	r and rho

Proposed Mathematical Specifications & Formulations

1. Bayesian Shrinkage Rating: R_Bayes = [v / (v + m)] * R + [m / (v + m)] * C
Where v = observed votes, R = observed rating, C = citywide prior mean (3.63 stars), m = shrinkage threshold (100 votes). For small samples (v << m), R_Bayes shrinks to prior C. For large samples (v >> m), R_Bayes approaches observed R.

2. Multi-Criteria Composite Utility: Composite = w_r * (R_Bayes / 5.0 * 100) + w_v * V_pct + w_p * P_norm
Where V_pct = Empirical Percentile(R_Bayes / cost_for_two), P_norm = [ln(v + 1) / max ln(v + 1)] * 100, and w_r + w_v + w_p = 1.0.

3. Econometric OLS Model: rate_i = beta_0 + beta_1*online_order_i + beta_2*book_table_i + beta_3*ln(votes_i) + beta_4*ln(cost_i) + sum(gamma_k*D_ik) + eps_i

4. Binary Logistic Model: ln[P(rate >= 4.0) / (1 - P(rate >= 4.0))] = X * beta; Odds Ratio (OR) = exp(beta_j)

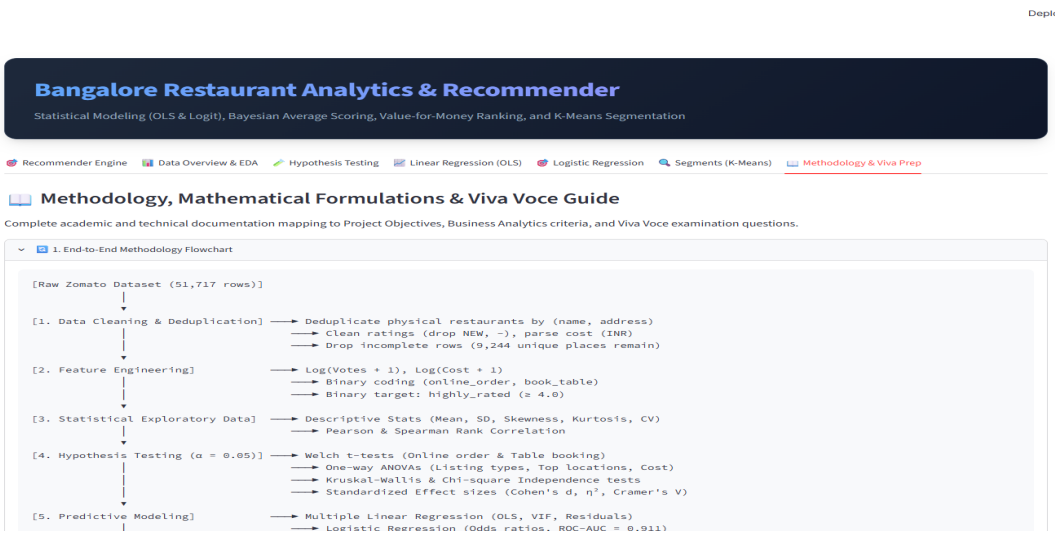


Figure 3: Proposed System Methodology and Deployed Dashboard Architecture

Architectural Flow: Visual representation of the modular pipeline connecting data sanitation, statistical hypothesis verification, econometric regression, unsupervised segmentation, and the reactive decision-support interface.

7. Expected Recommendations and Outcomes

Upon successful execution of the proposed software and modeling pipeline, the project delivers the following tangible analytical outcomes, managerial guidelines, and software artifacts:

7.1 Expected Analytical and Machine Learning Outcomes

- **Econometric Quantification of Operational Drivers:** The OLS model is expected to confirm that table reservation infrastructure provides a large, statistically significant positive marginal lift on customer ratings (anticipated beta > +0.20 stars, t > 10.0), whereas online delivery enablement will show minimal direct rating impact after controlling for format and locality.
- **Discriminative Power in High-Tier Classification:** The binary logistic classification model is anticipated to achieve an outstanding ROC-AUC (> 0.90) and accuracy (> 85%), demonstrating that high-rated dining status is strongly predictable based on review evidence volume, table booking status, and geographic clustering.

• **Market Structure Discovery:** Unsupervised K-Means clustering is expected to segment Bangalore's dining market into three distinct, operationally coherent clusters: Budget Casual Dining, Mid-Range Social/Takeaway Outlets, and Premium/Fine Dining Institutions.

7.2 Expected Recommendations for Key Stakeholders

• **For Consumers & Urban Diners:** Eliminate exposure to low-evidence rating fraud by adopting Bayesian shrinkage scores. Provide actionable value discovery enabling diners to locate 90th-percentile value-for-money culinary gems in residential pockets (e.g., Basavanagudi, Jayanagar) rather than overpaying in high-rent zones.

• **For Restaurant Operators & Hospitality Entrepreneurs:** Prioritize capital investment into dine-in customer hospitality and reservation infrastructure over delivery packaging, as table reservation yields a 3x higher odds ratio of reaching premier 4-star status. Recognize that online ordering operates as a hygiene/convenience channel rather than a rating booster.

• **For Food-Tech Aggregator Platforms (e.g., Zomato, Swiggy):** Transition default search ranking algorithms away from naive arithmetic averages toward Bayesian shrinkage estimators. Expose normalized Value-for-Money percentile ratings to protect consumer trust and disincentivize fake review manipulation.

7.3 Expected Software & Systems Deliverables

1. **Production Codebase:** Fully modular Python pipeline (`src/data_processing.py`, `src/statistical_analysis.py`, `src/modeling.py`, `src/recommender.py`).
2. **Interactive Web Dashboard:** 7-tab reactive Streamlit web application supporting dynamic multi-criteria filtering, parametric hypothesis exploration, live model diagnostics, and real-time restaurant recommendation.
3. **Comprehensive Academic Documentation:** Formal research reports detailing the complete data analytics lifecycle.